

Pioneer calls it a day

Pioneer will phase out all its TV-related business activities by the year 2010. It will however remain in the audio business.

TOOLBOX

sound stage behind the head and in combination with the front loudspeakers, to the sides. A central loudspeaker located behind or just above the middle of the screen is used to reinforce the centre of the sound image for dialogue of characters in close-up. A 7.1 system extends the 5.1 layout by adding two further loudspeakers – back left and back right. In 6.1 there is only a single centre back loudspeaker.

However, in most domestic situations setting up even a 5.1 system can be tricky. Placing the two rear speakers presents the greatest difficulty because the ideal position for these is just a little behind and above the listener, effectively placing them on stands in the middle of the room.

For a stable and accurate surround image it is important that the loudspeakers in a surround sound system are all from a matched set. It is inadvisable to try to build up a surround system by adding loudspeakers to an existing left/right stereo pair unless they are identical models.

A 5.1 system requires quite a lot of space, very symmetrical rooms should be avoided and hard reflective surfaces – the ceiling above and the floor in front of the listening position – should be damped. Floor reflections can be controlled using carpets while ceilings can be fitted with acoustic panels. It may also be desirable to fit acoustic damping to the wall behind the listening position.

While the single bass loudspeaker can be placed almost anywhere in the room, some positions will sound better than others and finding the right position is a matter of trial and error. Perhaps the ideal position is mid-way between the two front loudspeakers just below the screen. It is important to

balance the relative levels between the small satellite loudspeakers and the sub-woofer and 5.1 systems usually have a separate gain control for the sub-woofer. Placing loudspeakers at junctions between walls and floor or ceiling will boost bass, although this may not be an issue with small satellite speakers. Putting the sub-woofer in a corner may seem convenient, but it will result in a large bass boost and it may not be possible to balance the bass content using the bass gain control.

Active loudspeakers

Smaller 5.1 systems, particularly those designed for computer gaming, may have a five channel amplifier incorporated into the bass sub-woofer cabinet, eliminating the need for a separate amplifier. These can be reasonably effective in smaller rooms. At the other end of the scale, very high-end systems may also use wide frequency range active loudspeakers with larger cabinets. Sub-woofers may not be needed with such systems.

Program sources – aerial, cable or satellite?

Standard definition analogue television broadcasts are gradually giving way to DVB-T (Digital Video Broadcasts – Terrestrial) and existing televisions can be converted for digital reception with the addition of a digital tuner, often in a set top box. Even in major population centres reception of the digital terrestrial broadcast signals can be poor, the digital broadcasts do need a different aerial to that used for the analogue signals and you may need a high gain, very directional 'digital' aerial, possibly even fitted with a signal boosting amplifier. Using an aerial allows access to the 'free' broadcast digital channels. However it may be more convenient to subscribe to a satellite or cable service using a set top box, which should guarantee a strong signal. Even with a satellite or cable service it's not guaranteed that program reception won't be interrupted from time to time. Satellite reception is perhaps slightly more likely to drop the signal than a cable hook up, depending on the weather, the condition of the satellite network and sunspot activity.

HDTV is just starting to become available in India this year. Many program makers are changing to HD and Direct to Home (DTH) satellite packages are available from Dish TV, Reliance Communications BIG TV, Tata Sky, Sundirect DTH, Airtel Digital TV and the public sector DD Direct (Doordarshan). HD set top boxes from Dish TV cost between Rs. 3,300 and Rs. 4,000, around twice the price of standard definition. Cable services are also extremely popular in India and since a furore in 2001 over large price hikes, have been provided on a Conditional Access System (CAS). CAS allows consumers to choose and only pay for those channels they want to watch.

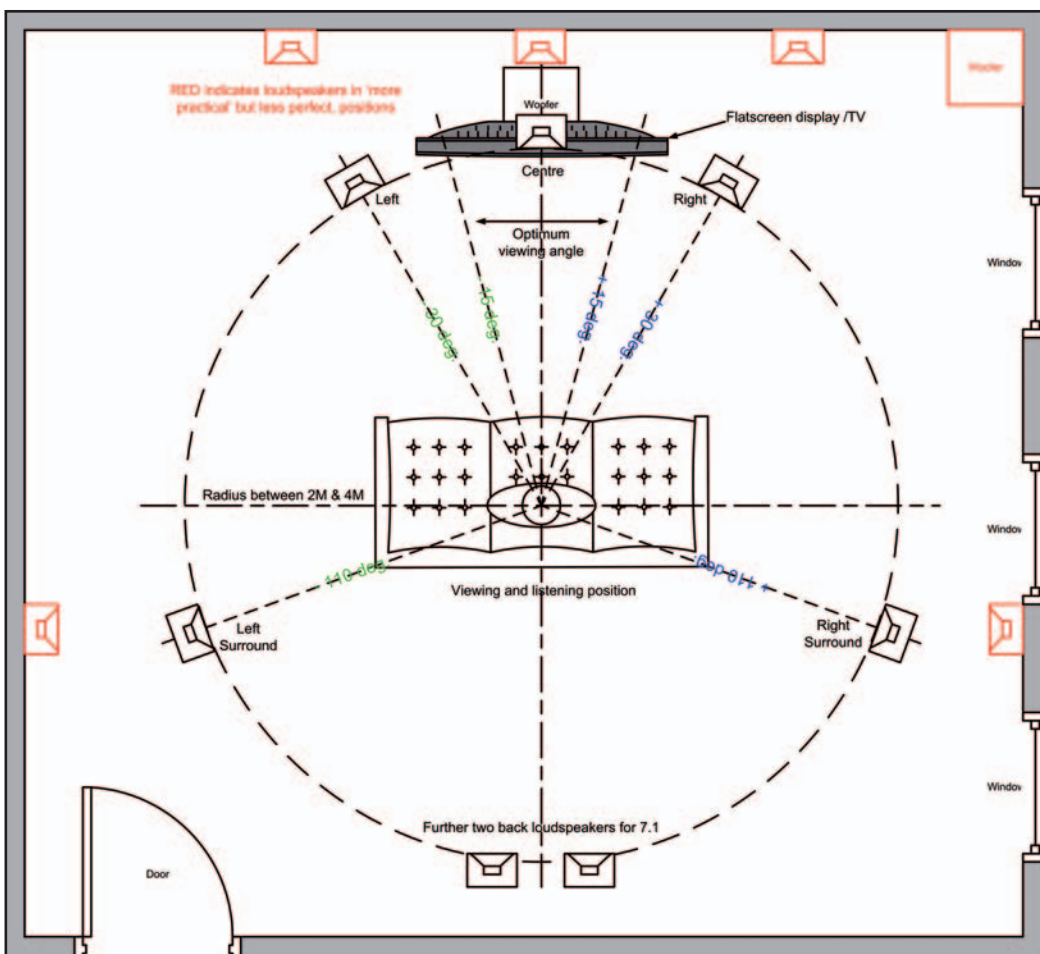


Diagram of a 5.1 audio setup in a typical room showing the possible inconvenience caused by placing the loudspeakers in the optimum positions